

Exercise electrodermal dynamics: persistence survives physiological challenges

The coherent aerobic subset contains 30 participants with wrist electrodermal activity (EDA), temperature and accelerometry. Six people are reserved, including both linked S11 recording segments; 24 people develop models in five whole-person folds. Source 4 Hz EDA is in microSiemens, temperature in Celsius, and accelerometer units are converted using the source 1/64 g convention.

Task and equation

Every five seconds after 120 seconds of observed history, the task predicts mean EDA over seconds 25–30 ahead. The development-selected reference is

$$\widehat{E}_{[t+25, t+30]} = \overline{E}_{[t-5, t]}.$$

The rule has no fitted parameters. Each sensor's own start timestamp and sampling rate determine alignment; recording breaks are never bridged. The authors shift dates by a random number of days exceeding one year for privacy, consistently across sensors; these are alignment coordinates, not measurement calendar dates. Signed finite EDA is retained as released rather than silently clipped or excluded.

Five hypotheses tested tonic relaxation, damped drift, preceding activity, thermal change and asymmetric rising/recovery dynamics. Every extension was fitted only on training people with whole-person validation and identical observed-history access. The apparent physiological coefficients failed to improve the strong persistence control.

Frozen results

Model	Development group error	Reserved group error
baseline_persistence	0.8210218	0.4901973
baseline_velocity	1.469895	1.123018
baseline_flexible	0.8902059	0.5420116
attempt_001_relaxation	0.8367453	0.500288
attempt_002_damped_drift	0.8598989	0.5246933
attempt_003_activity	0.8732695	0.5355587
attempt_004_thermal	0.8601812	0.5246942
attempt_005_asymmetry	0.8500905	0.5206393

Selected before confirmation: **baseline_persistence**. Primary error is mean absolute error in uS, averaged within each independent group and then equally across groups. All candidates are shown to preserve unfavorable results; a lower retrospective score does not change the selected reference.

Reserved participant-balanced MAE is 0.4902 uS for persistence, versus 1.1230 uS linear extrapolation and 0.5420 uS flexible prediction. This is useful bounded forecasting evidence and a negative result for the tested physiological extensions. It is not a new sweat-production law, a mental-stress classifier or a clinically validated wearable. Source exercise cadence changes and disconnections are preserved; the two source protocol versions within one study do not isolate temperature, autonomic activity and motion causally. Six held people warrant per-person error reporting, not narrow population confidence intervals.

Validation and reproducibility

After 120 s observed history, every 5 s predict a future 5 s EDA mean ending 30 s later. All EDA, temperature and accelerometer summaries are causal and aligned by native starts/sample rates. No held-person outcome calibration. Past 120 s EDA is permitted state, identical for every model; response coefficients and flexible transforms fit training people only.

One aerobic study with two protocol versions; no mental-stress or clinical diagnosis. Source device clocks are not necessarily calendar measurement dates, and manufacturer EDA calibration remains author-provided. All outcomes are now exposed to later agents. Claims of new validation require fresh data or an explicitly retrospective designation. Source study and processing are disclosed in SOURCE_AND_UNITS and PRIOR_ART; this is computational confirmation, not independent experimental replication.

`python run.py` verifies the allowlisted package and reproduces all saved predictions. `rules.json` contains exact coefficients and all learned transforms; `EQUATIONS.md` names each feature and equation. The adjacent native adapter reconstructs source-hash-verified observations and predictors; `task_spec.json` declares input, timing, calibration and grouping budgets. Future proposals may use different equations under the same contract, or seek review of a genuinely different measured task.

Source: [Causal electrodermal response during exercise; primary source/prior art](#). License recorded by source: ODC Attribution 1.0. Literature and semantics audited 2 October 2026.

The official version 1.0.1 dataset DOI is [10.13026/he0v-tf17](https://doi.org/10.13026/he0v-tf17). The two source recruitment stages use different aerobic schedules (Sxx/fxx); holding out participants does not establish transfer to an independently designed study.